DEPARTMENT OF MATHEMATICS

Coding Theory — Module 2

ASSIGNMENT

Part I *BSC Probability, Hamming Distance & Error Codes (Q1–Q5)*

Q1	**Binary Symmetric Channel — Probability | Hamming Distance & Error Capability**	**14M**
A	**A binary symmetric channel has error probability p = 0.03.** A word of length 9 is transmitted. **(i)** Determine the probability that exactly one error occurs. **(ii)** Determine the probability that exactly two errors occur. **(iii)** Determine the probability that exactly two errors occur and they are not consecutive.	7M
B	The following codewords form a code: `00000,    10110,    01101,    11011` Determine: **(a)** The minimum Hamming distance. **(b)** The error detection capability. **(c)** The error correction capability.	7M
Q2	**BSC Probability for a Given Word | Pairwise Hamming Distances**	**14M**
A	A binary symmetric channel has p = 0.04. The word **e = 10110011** is transmitted. **(i)** Find the probability of exactly one error. **(ii)** Find the probability of exactly three errors. **(iii)** Find the probability that exactly three errors occur such that no two are consecutive.	7M
B	The codewords of a code are: `11001,    10111,    01010,    00110` Determine: **(a)** All pairwise Hamming distances. **(b)** The minimum distance. **(c)** The error detection capability.	7M
Q3	**Word Transmission Probabilities — Error Positions | Encoding Function Analysis**	**14M**
A	A word of length 7 is transmitted through a binary symmetric channel with p = 0.08. **(i)** Determine the probability of no error. **(ii)** Determine the probability of at least one error. **(iii)** Determine the probability that errors occur exactly at positions 3 and 6 only.	7M

B	A (3, 8) encoding function **e : $B^3 \to B^8$** is defined by:
	`e(000)  =  0000 0000`
	`e(001)  =  1011 1000`
	`e(010)  =  0100 1101`
		7M

	`e(011) = 1111 0101`
	`e(100) = 1010 0100`
	`e(101) = 0101 1001`
	`e(110) = 0010 1100`
	`e(111) = 1101 0001`
	Determine:
	(a) How many errors will e detect?
	(b) How many errors will e correct?	

Q4	**Probability of Errors in Specific Positions | Minimum Distance of Encoding Map**	**14M**
A	A word of length 8 is transmitted over a binary symmetric channel with p = 0.1. (i) Determine the probability that exactly two errors occur in even positions only. (ii) Obtain the probability of exactly two errors anywhere. (iii) Determine the probability that at least 3 errors occur.	**7M**

B	An encoding function $\mathbf{E : Z_2^3 \to Z_2^6}$ is defined by:
	`000 → 000000`
	`001 → 001011`
	`010 → 010101`
	`011 → 011110`
	`100 → 100110`
	`101 → 101101`
	`110 → 110011`
	`111 → 111000`
	Determine:
	(a) The minimum Hamming distance between any two codewords.
	(b) The number of errors that can be detected.
	(c) The number of errors that can be corrected.
		7M

Q5	**BSC with p = 0.02 | Minimum Distance & Error Capability from Codewords**	**14M**
A	A binary symmetric channel has probability p = 0.02 of incorrect transmission. The word **e = 101011001** is transmitted. (i) Find the probability of exactly one error. (ii) Find the probability that exactly two errors occur. (iii) Find the probability of three errors such that no two are adjacent.	**7M**
B	The codewords are: `000000, 001110, 110001, 111111` Determine: (a) All pairwise Hamming distances between codewords. (b) The minimum distance. (c) The error detection and correction capability.	**7M**

Q6 Generator Matrix — Encoding, Parity-Check Matrix & Decoding — 14M

▸ 14M

Given the generator matrix for the encoding function $\mathbf{E} : \mathbb{Z}_2^3 \to \mathbb{Z}_2^6$:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

(i) Determine the codewords for 010 and 101.

(ii) Determine the associated parity-check matrix H.

`010101` and `101110`.

Q7 Encoding Function $\mathbf{E} : \mathbb{Z}_2^2 \to \mathbb{Z}_2^5$ — Codewords, H-Matrix & Decoding — 14M

▸ 14M

An encoding function $\mathbf{E} : \mathbb{Z}_2^2 \to \mathbb{Z}_2^5$ is given by the generator matrix:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

(i) Determine all the codewords.

(ii) Obtain the associated parity-check matrix H.

`10101` and `01110`.

Q8 Generator Matrix $\mathbf{E} : \mathbb{Z}_2^3 \to \mathbb{Z}_2^6$ — Codewords, H-Matrix & Decoding — 14M

▸ 14M

The generator matrix for an encoding function $\mathbf{E} : \mathbb{Z}_2^3 \to \mathbb{Z}_2^6$ is:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

(i) Determine the codewords assigned to 101 and 011.

(ii) Obtain the associated parity-check matrix H.

`101011` and `011101`.

Q9 Syndrome Decoding Table — Group Code from Generator Matrix — 14M

▸ 14M

Construct a complete decoding table (with syndromes) for the group code generated by:

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

for the encoding function $\mathbf{E} : \mathbb{Z}_2^2 \to \mathbb{Z}_2^5$.

`11010`, `10110`, `01001`, `00111`.

Q10 Generator Matrix — Encoding, Parity-Check Matrix & Decoding — 14M

▸ 14M

For the generator matrix of the encoding function $\mathbf{E} : \mathbb{Z}_2^3 \to \mathbb{Z}_2^6$:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

(i) Encode 100 and 011.

(ii) Find the associated parity-check matrix H.

`100101` and `011011`.

— End of Assignment —